AFFINE DIAGONAL MAKES THE COHERATOR AN EQUIVALENCE ON A QUASI-COMPACT SPACE

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Editorial note. Counted span is the author's own proof environment, source lines 2702-2711, immediately following the statement at 2688-2700.

Editorial note. Hypothesis-verification pattern: the span's own content is the affine-permanence check; the equivalence conclusion is the terminal criterion.

Allowed standard prerequisites (editorial).

- The abstract criterion. Lemma lemma-argument-proves of this chapter, cited at proof lines 2709-2711. Terminal input: the criterion under which the comparison functor is an equivalence with quasi-inverse RQ_X.
- Affine pushforward vanishing input. Lemma lemma-affine-pushforward of this chapter, cited at proof lines 2708-2709. Supplies the cohomological consequence once maps from affines into quasi-compact opens are affine.

Source reference. Original equation 08F1: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 509–509. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08GY: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2339–2339. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09TG: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2490–2490. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08GB: focused-seeds/geometry/S9-perf-quasi-compact-affine-diagonal/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03BS: focused-seeds/geometry/S9-perf-quasi-compact-affine-diagonal/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

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Proposition 0.1. *Let S be a scheme. Let X be a quasi-compact algebraic space over S with affine diagonal over $\mathbf{Z}$ (as in Properties of Spaces, Definition 03BS). Then the functor (08F1)*

$$D(QCoh(\mathcal{O}_X)) \longrightarrow D_{QCoh}(\mathcal{O}_X)$$

is an equivalence with quasi-inverse given by RQ_X.

Proof. Let $V \to W$ be an étale morphism with V affine and W a quasi-compact open subspace of X. Then the morphism $V \to W$ is affine as W has affine diagonal over $\mathbf{Z}$ and V is affine (Morphisms of Spaces, Lemma 08GB). Lemma 08GY then guarantees that the assumption of Lemma 09TG holds. Hence we conclude. □